EXPERIMENT 2

Develop a multi-layer perceptron (MLP) for a simple classification task. Experiment with different numbers of hidden layers and activation functions, and evaluate the model's performance using accuracy and loss.

```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, log_loss

iris = load_iris()

X = iris.data
y = iris.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

models = {
    "1 Hidden Layer": (10,),
    "2 Hidden Layers": (10, 10),
    "3 Hidden Layers": (20, 10, 5)
}

for name, layers in models.items():
    model = MLPClassifier(
        hidden_layer_sizes=layers,
        activation="relu",
        solver="adam",
        max_iter=1000,
        random_state=42
    )

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)

    accuracy = accuracy_score(y_test, y_pred)
    loss = log_loss(y_test, y_prob)
```

```
    print("\nModel:", name)
    print("Accuracy:", accuracy)
    print("Loss:", loss)
```

```
Model: 1 Hidden Layer
Accuracy: 1.0
Loss: 0.059564389893335644

Model: 2 Hidden Layers
Accuracy: 0.9666666666666667
Loss: 0.05663264435333653

Model: 3 Hidden Layers
Accuracy: 1.0
Loss: 0.017622281019418453
```